

James C. Davis

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RESEARCH THEME

My research develops empirical and engineering foundations for trustworthy software systems. I use evidence from experiments, mining software repositories, and real deployments to guide the design, verification, and security of software and its supply chains. By integrating empirical study with practical tool building, my group strengthens the reliability and resilience of the software that underpins modern computing.

EDUCATION

Ph.D, Computer Science and Applications <i>Virginia Tech, Blacksburg, VA</i>	2015–2020
B.Sc. Computer Science, B.Sc. Mathematics <i>Clarkson University, Potsdam, NY</i>	2008–2012

PROFESSIONAL EXPERIENCE

Assistant Professor <i>Purdue University — Electrical and Computer Engineering</i>	Fall 2020–present
Intern, Microsoft Research (RiSE group: Cloud Security) <i>Microsoft Research, Redmond, WA — Mentored by Patrice Godefroid</i>	Summer 2019
Intern, IBM Research (Storage) <i>IBM Research, Almaden, CA — Mentored by Deepavali Bhagwat</i>	Summer 2018
Graduate Research Assistant <i>Virginia Tech — Advised by Dongyoon Lee</i>	2016–2020
Software Engineer, IBM (GPFS) <i>IBM, Poughkeepsie, NY</i>	2012–2017

EXTERNAL GRANTS

TOTAL: \$5,262,934. TOTAL AS PI: \$2,576,026. MY TOTAL SHARE: \$3,391,251.¹

CLOSELY RELATED TO PROPOSED RESEARCH

[G-1] NSF #2541917: CAREER: PTM-SEER: Software Engineering Foundations for Re-Using Pre-Trained Neural Models
PI
US National Science Foundation
2026–2031. \$687,140; my share: \$687,140.
Other personnel: Sole PI

¹Calculation: Total is the sum of all awards to Purdue on which I have been PI or Co-PI. Total as PI is the sum of all awards to Purdue on which I have been the PI — not all of these funds are managed by me. My total share is the sum of funds I have managed across all awards.

- [G-2] **NSF #2526621: Collaborative Research: SaTC 2.0: RES: AIGIS: Securing the Deep Learning Model Supply Chain**
Purdue PI (Columbia University is lead)
US National Science Foundation
2026–2030. Total: \$1,220,000; Purdue: \$410,000; my share: \$410,000.
Other personnel: NSF #2526620: Junfeng Yang (Computer Science, Columbia University); NSF #2526622: Vasileios Kemerlis (Computer Science, Brown University)
- [G-3] **NSF #2504445: OAC Core: Cyberinfrastructure for Multi-Stream Architectures Applied to Computer Vision: Efficiency via Co-Design of Network Architectures and Framework Operators**
Co-PI
US National Science Foundation
2025–2028. \$660,000; my share: \$330,000.
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)
- [G-4] **Rolls Royce: Securing Software Implementations through System Fuzz Testing and Modular Formal Methods**
PI
Contract with Rolls Royce
2025–2026. \$181,447; my share: \$90,723.5.
Other personnel: Co-PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-5] **AutoUP: Automated Unit Proofing**
PI
OpenAI — Cybersecurity Grant Program
2025. \$5,000 (API credits); my share: \$5,000 (API credits).
Other personnel: Sole PI
- [G-6] **NSF #2343596: Collaborative Research: EAGER: CET: Exploring The Risks and Rewards of Large Language Models in Enabling Energy-Efficient Data Center Software Infrastructure**
Purdue PI (Loyola University Chicago is lead)
US National Science Foundation
2024–2026. Total: \$297,923; Purdue: \$149,976; my share: \$74,988.
Other personnel: Co-PI: Yung-Hsiang Lu (Electrical & Computer Engineering); NSF #2343595: George K. Thiruvathukal (Computer Science, Loyola University Chicago)
- [G-7] **Rolls Royce: Facilitating Effective Dynamic Analysis of Embedded Software**
Co-PI
Contract with Rolls Royce
2024–2025. \$150,000; my share: \$75,000.
Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)

- [G-8] **Rolls Royce: Dynamic Security Analysis of Embedded Software Systems**
Co-PI
Contract with Rolls Royce
2023-2024. \$150,000; my share: \$75,000.
Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-9] **Cisco: Trustworthy Re-use of Pre-Trained Neural Networks**
PI
Contract with Cisco
2022-2023. \$179,237; my share: \$89,618.5.
Other personnel: Co-PI: Yung-Hsiang Lu (Electrical & Computer Engineering)
- [G-10] **Unrestricted Gift: Machine Learning Reproducibility**
PI
Google, LLC
2022. \$80,000; my share: \$80,000.
Other personnel: Sole PI
- [G-11] **NSF #2107230: Collaborative Research: OAC Core: Advancing Low-Power Computer Vision at the Edge**
Co-PI (Purdue University is lead)
US National Science Foundation
2021-2024. Total: \$508,000; Purdue: \$258,000; my share: \$129,000.
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering); NSF #2107020: George K. Thiruvathukal (Computer Science, Loyola University Chicago)
- [G-12] **Rolls Royce: Dynamic Analysis of Embedded Firmware**
Co-PI
Contract with Rolls Royce
2021-2022. \$175,000; my share: \$87,500.
Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-13] **Unrestricted Gift: Research on Machine Learning Reproducibility**
Co-PI
Google, LLC
2020. \$100,000 (\$80,000 + \$20,000); my share: \$100,000 (\$80,000 + \$20,000).
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)

OTHER EXTERNAL RESEARCH SUPPORT

- [G-14] **NSF #2452533: RFE: Research: Developing and Piloting a Prompt Engineering Competency Framework for Software Engineering Education**
Co-PI
US National Science Foundation
2025-2028. \$350,000; my share: \$65,716.
Other personnel: PI: Kirsten Davis (Engineering Education)

- [G-15] **NSF #2537308: Collaborative Research: Planning: CROSS: Building a Community aROund Securing the Research Software Supply Chain**
PI (Purdue University is lead)
US National Science Foundation
2025–2027. Total: \$200,000; Purdue: \$105,113; my share: \$82,213.
Other personnel: Co-PI: Alexandra Harris-Watson (Psychological Sciences); NSF #2537310: Jeffrey C. Carver (Computer Science, University of Alabama Tuscaloosa); NSF #2537309: George K. Thiruvathukal (Computer Science, Loyola University Chicago)
- [G-16] **Qualcomm Innovation Fellowship**
PI
Qualcomm, Inc.
2025–2026. \$100,000; my share: \$100,000.
Other personnel: Project Supervisor of winning team: P. Amusuo, D. Anandayuvraj
- [G-17] **Unrestricted Gift: Typosquat Detection in Open-Source Ecosystems**
PI
Socket, Inc.
2025. \$20,000; my share: \$20,000.
Other personnel: Sole PI
- [G-18] **NSF #2512797: Travel: NSF Student Travel Grant for 2025 International Conference on Software Engineering**
PI
US National Science Foundation
2024-2025. \$24,000; my share: \$24,000.
Other personnel: Sole PI
- [G-19] **Unrestricted Gift: Improving OSS Supply Chain Security by Promoting Software Signing**
Co-PI
Google, LLC
2023. \$200,000; my share: \$100,000.
Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)
- [G-20] **Efficient Computer Vision for Edge Devices**
Co-PI
Contract with Cisco
2023-2024. \$179,941; my share: \$89,970.5.
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)
- [G-21] **NSF #2135156: Collaborative Research: SaTC: CORE: Small: Improving Sanitization and Avoiding Denial of Service Through Correct and Safe Regexes**
PI (Purdue University is lead)
US National Science Foundation
2022–2025. Total: \$548,000; Purdue: \$274,000; my share: \$274,000.
Other personnel: NSF #2135157: Dongyoon Lee (Computer Science, SUNY at Stony Brook)

[G-22] **NSF #2229703: POSE: Phase I: Scoping An Open-Source Ecosystem Around Proactive Software Supply Chain Monitoring**

Co-PI

US National Science Foundation

2022–2023. \$300,000; my share: \$150,000.

Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)

[G-23] **Cisco: Monitor and manage security risks in software supply chains with Sigstore**

Co-PI

Contract with Cisco

2022–2023. \$184,536; my share: \$92,268.

Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)

REFEREED CONFERENCE PUBLICATIONS (FULL PAPERS)² These venues are CORE2026 rank A or A*.³

- [C-1] P.C. Amusuo, R. Calvo, D. Anandayuvraj, T. Le Lievre, K. Kolyakov, E. Jorgensen, A. Machiry, and **J.C. Davis**. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification*. Proceedings of the 33rd ACM Conference on Computer and Communications Security (CCS'26). 15 pages.
- [C-2] C. Okafor, **J.C. Davis**, and S. Torres-Arias. *Context-Aware Trust Verification for Identity-Based Software Signing*. Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE'26). 21.2% acceptance rate (265/1,236). 13 pages.
- [C-3] K.G. Kalu, S. Okorafor, T. Singla, S. Chen, S. Torres-Arias, and **J.C. Davis**. *Why Johnny Adopts Identity-Based Software Signing: A Usability Case Study of Sigstore*. Proceedings of the 35th USENIX Security Symposium (USENIX Security'26). 13.2% acceptance rate (364/2,754). 20 pages.
- [C-4] Y. Liu, B. Çakar, A. Agrawal, M. Seo, **J.C. Davis**, and Dongyoon Lee. *Regular Expression Denial of Service Induced by Backreferences*. Proceedings of the 35th USENIX Security Symposium (USENIX Security'26). 13.2% acceptance rate (364/2,754). 20 pages.
- [C-5] P. Jajal, N.J. Eliopoulos, B.S.H. Chou, G.K. Thiruvathukal, Y.H. Lu, and **J.C. Davis**. *AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition — Findings Track (2026). 35.3% acceptance rate ((4,090+1,600)/16,092). 11 pages.
- [C-6] P. Jajal*, N. Eliopoulos*, B.S.H. Chou, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition — Findings Track (2026). 35.3% acceptance rate ((4,090+1,600)/16,092). 12 pages.

²Here and elsewhere, my research mentees are underlined. These are students whose work I supervised during the research project. My name is given in **bold**. In these publication venues, the final author (“anchor author”) provides the primary intellectual supervision of the work.

³In my research areas, top-tier conferences are the most prestigious venues for publication, with some journals considered comparable. The ICORE rankings are an international ranking system of conferences in which A* venues are “best” and A venues are “reputable” venues; for either category, the peer reviewers are recognized experts in the field. Acceptance rates of ~10–25% across 100+ submissions are a good indicator of conference quality.

- [C-7] A. Rozema and **J.C. Davis**. *Anti-Phishing Training (Still) Does Not Work: A Reproduction of Phishing Training Inefficacy Grounded in the NIST Phish Scale*. Proceedings of the ACM Web Conference 2026 (WWW'26). 20.1% acceptance rate (676/3,370). 12 pages.
- [C-8] W. Jiang, B. Çakar, M. Lysenko, and **J.C. Davis**. *ConfuGuard: Using Metadata to Detect Active and Stealthy Package Confusion Attacks Accurately and at Scale*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.
- [C-9] D. Anandayuvraj, T. Singla, Z. Hammadeh, A. Lund, A. Holloway, and **J.C. Davis**. *Learning From Software Failures: A Case Study at a National Space Research Center*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.
- [C-10] P.C. Amusuo, O. Cochell, T. Le Lievre, P.V. Patil, A. Machiry, and **J.C. Davis**. *Do Unit Proofs Work? An Empirical Study of Compositional Bounded Model Checking for Memory Safety Verification*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.
- [C-11] B.S.H. Chou, P. Jajal, N.J. Eliopoulos, **J.C. Davis**, G.K. Thiruvathukal, K.Y.J. Yun, and Y.H. Lu. *LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection*. Proceedings of the Fourteenth International Conference on Learning Representations (ICLR'26). 28% acceptance rate (5,000/18,000). 22 pages.
- [C-12] A. Kellas, N. Christou, W. Jiang, P. Li, L. Simon, Y. David, V. Kemerlis, **J.C. Davis**, and J. Yang. *PickleBall: Secure Deserialization of Pickle-based Machine Learning Models*. Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS'25). 13.9% acceptance rate (316/2,278). 16 pages. *Best Artifact Award*.
- [C-13] P.C. Amusuo, K.A. Robinson, T. Singla, H. Peng, A. Machiry, S. Torres-Arias, L. Simon, and **J.C. Davis**. *ZTD_{JAVA}: Mitigating Software Supply Chain Vulnerabilities via Zero-Trust Dependencies*. Proceedings of the 47th International Conference on Software Engineering (ICSE'25). 24% acceptance rate (248/1031). 13 pages.
- [C-14] B.S.H. Chou, P. Jajal, N.J. Eliopoulos, T. Nadolsky, C.Y. Yang, N. Ravi, **J.C. Davis**, K.Y.J. Yun, and Y.H. Lu. *A Musician's Muse: Detecting Performance Errors with Transformers*. Proceedings of the 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25). 23% acceptance rate (3,032/12,957). 11 pages.
- [C-15] S.R. Tanksalkar, S. Muralee, D.M.S.H. Danduri, P. Amusuo, A. Bianchi, **J.C. Davis**, and A. Machiry. *LEMIX: Enabling Testing of Embedded Applications as Linux Applications*. Proceedings of the 34th USENIX Security Symposium (USENIX Security'25). 17.1% acceptance rate (407/2,385). 30 pages.
- [C-16] K. Kalu, T. Singla, C. Okafor, S. Torres-Arias, and **J.C. Davis**. *An Industry Interview Study of Software Signing for Supply Chain Security*. Proceedings of the 34th USENIX Security Symposium (USENIX Security'25). 17.1% acceptance rate (407/2,385). 18 pages.
- [C-17] M. Shen, A. Pillai, B.A. Yuan, **J.C. Davis**, and A. Machiry. *Finding 709 Defects in 258 Projects: An Experience Report on Applying CodeQL to Open-Source Embedded Software (Experience Paper)*. Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'25). 21% acceptance rate (115/553). 22 pages.
- [C-18] M.H.M. Bhuiyan*, B. Çakar*, E. Burmane, **J.C. Davis**, and C.A. Staicu. *SoK: A Literature and Engineering Review of Regular Expression Denial of Service*. Proceedings of the 20th ACM ASIA Conference on Computer and Communications Security (AsiaCCS) Cycle 2 (AsiaCCS'25). 20% acceptance rate (53/269). 22 pages.

- [C-19] N. Eliopoulos, P. Jajal, **J.C. Davis**, G. Liu, G.K. Thiruvathukal, and Y.H. Lu. *Pruning One More Token is Enough: Leveraging Latency-Workload Non-Linearities for Vision Transformers on the Edge*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'25). 37.8% acceptance rate (930/2,458). 12 pages.
- [C-20] P. Jajal, N. Eliopoulos, B. Chou, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Token Turing Machines are Efficient Vision Models*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'25). 37.8% acceptance rate (930/2,458). 12 pages.
- [C-21] D. Anandayuvraj, M. Campbell, A. Tewari, and **J.C. Davis**. *FAIL: Analyzing Software Failures from the News Using LLMs*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'24). 26% acceptance rate (154/587). 13 pages.
- [C-22] P. Jajal, W. Jiang, A. Tewari, E. Kocinare, J. Woo, A. Sarraf, Y.H. Lu, G.K. Thiruvathukal, and **J.C. Davis**. *Interoperability in Deep Learning: A User Survey and Failure Analysis of ONNX Model Converters*. Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'24). 21% acceptance rate (143/694). 13 pages.
- [C-23] J. Jones, W. Jiang, N. Synovic, G.K. Thiruvathukal, and **J.C. Davis**. *What do we know about Hugging Face? A systematic literature review and quantitative validation of qualitative claims*. Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM'24). 24% acceptance rate (34/139). 12 pages.
- [C-24] L. Franke, H. Liang, S. Farzanehpour, A. Brantly, **J.C. Davis**, and C. Brown. *An Exploratory Mixed-methods Study on General Data Protection Regulation (GDPR) Compliance in Open-Source Software*. Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM'24). 24% acceptance rate (34/139). 12 pages.
- [C-25] J. Chen, D. Anandayuvraj, **J.C. Davis**, and S. Rahaman. *On the Contents and Utility of IoT Cybersecurity Guidelines*. Proceedings of the ACM on Software Engineering (PACMSE), Issue FSE 2024 (FSE'24). 26% acceptance rate (121/474). 24 pages.
- [C-26] T.R. Schorlemmer, K.G. Kalu, L. Chigges, K.M. Ko, E.A.M.A. Ishgair, S. Bagchi, S. Torres-Arias, and **J.C. Davis**. *Signing in Four Public Software Package Registries: Quantity, Quality, and Influencing Factors*. Proceedings of the 45th IEEE Symposium on Security and Privacy (IEEE S&P'24). 18% acceptance rate (261/1463). 16 pages.
- [C-27] W. Maxam and **J.C. Davis**. *An Interview Study on Third-Party Cyber Threat Hunting Processes in the U.S. Department of Homeland Security*. Proceedings of the 33rd USENIX Security Symposium (USENIX Security'24). 18% acceptance rate (382/2176). 18 pages.
- [C-28] W. Jiang, J. Yasmin, J. Jones, N. Synovic, J. Kuo, N. Bielanski, Y. Tian, G.K. Thiruvathukal, and **J.C. Davis**. *PeaTMOSS: A Dataset and Initial Analysis of Pre-Trained Models in Open-Source Software*. Proceedings of the 21st Annual Conference on Mining Software Repositories (MSR'24). 29% acceptance rate (42/146). 13 pages.
- [C-29] W. Jiang, N. Synovic, M. Hyatt, T.R. Schorlemmer, R. Sethi, Y.H. Lu, G.K. Thiruvathukal, and **J.C. Davis**. *An Empirical Study of Pre-Trained Model Reuse in the Hugging Face Deep Learning Model Registry*. Proceedings of the ACM/IEEE 45th International Conference on Software Engineering (ICSE'23). 26% acceptance rate (208/796). 13 pages.
- [C-30] P.C. Amusuo, R.A.C. Méndez, Z. Xu, A. Machiry, and **J.C. Davis**. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23). 21% acceptance rate (134/629). 13 pages.

- [C-31] S.A. Hassan, Z. Aamir, D. Lee, **J.C. Davis**, and F. Servant. *Improving Developers' Understanding of Regex Denial of Service Tools through Anti-Patterns and Fix Strategies*. Proceedings of the 44th IEEE Symposium on Security and Privacy (IEEE S&P'23). 17.1% acceptance rate (196/1,147). 18 pages.
- [C-32] A. Goel, C. Tung, N. Eliopoulos, X. Hu, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Directed Acyclic Graph-based Neural Networks for Tunable Low-Power Computer Vision*. Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED'22). 24% acceptance rate (23/96). 6 pages.
- [C-33] E. Barlas, X. Du, and **J.C. Davis**. *Exploiting Input Sanitization for Regex Denial of Service*. Proceedings of the ACM/IEEE 44th International Conference on Software Engineering (ICSE'22). 26% acceptance rate (197/751). 13 pages.
- [C-34] Q. Xu, **J.C. Davis**, Y.C. Hu, and A. Jindal. *An Empirical Study on the Impact of Parameters on Mobile App Energy Usage*. Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'22). 36% acceptance rate (72/199). 12 pages.
- [C-35] A. Goel, C. Tung, X. Hu, H. Wang, **J.C. Davis**, Thiruvathukal, and Lu. *Low-Power Multi-Camera Object Re-Identification using Hierarchical Neural Networks*. Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED'21). 26.1% acceptance rate (23/88). 6 pages.
- [C-36] **J.C. Davis**, F. Servant, and D. Lee. *Using Selective Memoization to Defeat Regular Expression Denial of Service (ReDoS)*. Proceedings of the 42nd IEEE Symposium on Security and Privacy (IEEE S&P'21). 12% acceptance rate (115/952). 17 pages.
- [C-37] A. Cha, E. Wittern, G. Baudart, **J.C. Davis**, L. Mandel, and J. Laredo. *A Principled Approach to GraphQL Query Cost Analysis*. Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'20). 28% acceptance rate (101/360). 12 pages. *ACM Distinguished Paper Award*.
- [C-38] L. Rupprecht, **J.C. Davis**, C. Arnold, Y. Gur, and D. Bhagwat. *Improving Reproducibility of Data Science Pipelines through Transparent Provenance Capture*. Proceedings of the 46th International Conference on Very Large Data Bases (VLDB'20–Industry track). 20.6% acceptance rate (241/1,172). 15 pages.
- [C-39] **J.C. Davis**, D. Moyer, A. Kazerouni, and D. Lee. *Testing Regex Generalizability And Its Implications: A Large-Scale Many-Language Measurement Study*. Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE'19). 21% acceptance rate (91/435). 13 pages.
- [C-40] L. Michael, J. Donohue, **J.C. Davis**, D. Lee, and F. Servant. *Regexes are Hard: Decision-making, Difficulties, and Risks in Programming Regular Expressions*. Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE'19). 21% acceptance rate (91/435). 12 pages. *ACM Distinguished Paper Award*.
- [C-41] **J.C. Davis**, L. Michael, C. Coghlan, F. Servant, and D. Lee. *Why Aren't Regular Expressions a Lingua Franca? An Empirical Study on the Re-use and Portability of Regular Expressions*. Proceedings of the 27th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'19). 24% acceptance rate (97/371). 12 pages.
- [C-42] E. Wittern, A. Cha, **J.C. Davis**, G. Baudart, L. Mandel. *An Empirical Study of GraphQL Schemas*. Proceedings of the 17th International Conference on Service-Oriented Computing (ICSOC'19). 15% acceptance rate (28/183). 16 pages.
- [C-43] X. Fu, T. Ghaffar, **J.C. Davis**, and D. Lee. *EdgeWise: A Better Stream Processing Engine for the Edge*. 2019 USENIX Annual Technical Conference (USENIX ATC'19). 20% acceptance rate (71/356). 17 pages.

- [C-44] **J.C. Davis**, C. Coghlan, F. Servant, and D. Lee. *The Impact of Regular Expression Denial of Service (REDOS) in Practice: an Empirical Study at the Ecosystem Scale*. Proceedings of the 26th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'18). 21% acceptance rate (61/289). 11 pages. *ACM Distinguished Paper Award*.
- [C-45] **J.C. Davis**, E.R. Williamson, and D. Lee. *A Sense of Time for JavaScript and Node.js: First-Class Timeouts as a Cure for Event Handler Poisoning*. Proceedings of the 27th USENIX Security Symposium (USENIX Security'18). 19% acceptance rate (100/520). 17 pages.
- [C-46] **J.C. Davis**, A. Thekumparampil, and D. Lee. *Node.fz: Fuzzing the Server-Side Event-Driven Architecture*. Proceedings of the Twelfth European Conference on Computer Systems (EuroSys'17). 21% acceptance rate (41/200). 16 pages.

REFEREED JOURNAL ARTICLES⁴

- [J-1] H. Gao, M. Zahedi, W. Jiang, H.Y. Lin, **J.C. Davis**, and C. Treude. *AI Safety in the Eyes of the Downstream Developer: A First Look at Concerns, Practices, and Challenges*. Empirical Software Engineering, 2026 (EMSE'26). 44 pages.
- [J-2] J. Yasmin, W. Jiang, **J.C. Davis**, and Y. Tian. *Software Dependencies 2.0: An Empirical Study of Reuse and Integration of Pre-Trained Models in Open-Source Projects*. Empirical Software Engineering, 2026 (EMSE'26). 61 pages.
- [J-3] H. Peng, A. Gupte, R. Hasler, N. Eliopoulos, C. Ho, R. Mantri, L. Deng, K. Läufer, G.K. Thiruvathukal, and **J.C. Davis**. *SysLLMatic: Large Language Models are Software System Optimizers*. Journal of Systems and Software, 2026 (JSS'26). 30 pages.
- [J-4] W. Jiang, M. Kim, C. Cheung, H. Kim, G.K. Thiruvathukal, and **J.C. Davis**. *"I see models being a whole other thing": An Empirical Study of Pre-Trained Model Naming Conventions and A Tool for Enhancing Naming Consistency*. Empirical Software Engineering, 2025 (EMSE'25). 67 pages.
- [J-5] G. Cramer, W. Maxam, and **J.C. Davis**. *Engineering Patterns for Trust and Safety on Social Media Platforms: A Case Study of Mastodon and Diaspora*. Journal of Systems and Software, 2025 (JSS'25). 22 pages.
- [J-6] D. Özkan, K. Davis, **J.C. Davis**, J. Deters, and H. Murzi. *Fostering Systems Thinking through Engineering Study Abroad Programs*. European Journal of Engineering Education, 2024 (EJEE'24). 26 pages.
- [J-7] W. Jiang, V. Banna, N. Vivek, A. Goel, N. Synovic, G.K. Thiruvathukal, and **J.C. Davis**. *Challenges and Practices of Deep Learning Model Reengineering: A Case Study on Computer Vision*. Empirical Software Engineering, 2024 (EMSE'24). 63 pages.
- [J-8] K. Davis, J. Deters, D. Özkan, **J.C. Davis**, and H. Murzi. *Applying Experiential Learning Theory to Understand Study Abroad Leaders' Experiences Using Real-Time Perspectives*. Frontiers: The Interdisciplinary Journal of Study Abroad, Vol. 34, No. 2, 2022 (Frontiers'22). 31 pages.

⁴In the field of software engineering research, four journals are considered peers of the top conferences, as evidenced by their inclusion in "journal first" tracks at those conferences. Those journals are: ACM TOSEM, IEEE TSE, Springer EMSE, and Elsevier JSS.

- [J-9] S. Herbold, A. Trautsch, B. Ledel, A. Aghamohammadi, T.A. Ghaleb, K.K. Chahal, T. Bossenmaier, B. Nagaria, P. Makedonski, M.N. Ahmadabadi, K. Szabados, H. Spieker, M. Madeja, N. Hoy, V. Lenarduzzi, S. Wang, G. Rodriguez-Perez, R. Colomo-Palacios, R. Verdecchia, P. Singh, Y. Qin, D. Chakroborti, W. Davis, V. Walunj, H. Wu, D. Marcilio, O. Alam, A. Aldaej, I. Amit, B. Turhan, S. Eismann, A.K. Wickert, I. Malavolta, M. Sulir, F. Fard, A.Z. Henley, S. Kourtzanidis, E. Tüzün, C. Treude, S.M. Shamasbi, I. Pashchenko, M. Wyrich, **J.C. Davis**, A. Serebrenik, E. Albrecht, E.U. Aktas, D. Strüber, and J. Erbel. *A Fine-grained Data Set and Analysis of Tangling in Bug Fixing Commits*. Empirical Software Engineering, 2022 (EMSE’22). 55 pages.
- [J-10] A. Kazerouni, **J.C. Davis**, A. Basak, C. Shaffer, F. Servant, and S. Edwards. *Fast and Accurate Incremental Feedback for Students’ Software Tests Using Selective Mutation Analysis*. Journal of Systems and Software, 2021 (JSS’21). 22 pages.
- [J-11] D. Özkan, K. Davis, **J.C. Davis**, M. James, H. Murzi, and D. Knight. *Expectations and Experiences of Short-Term Study Abroad Leadership Teams*. Journal of International Engineering Education, 2020 (JIEE’20). 34 pages.

BOOKS AND BOOK CHAPTERS

- [B-1] **J.C. Davis**. (2026). *Model-Based Agentic Software Engineering*. 441 pages. Book manuscript. <https://davisjam.github.io/model-based-agentic-software-engineering/book/index.html>
- [B-2] **J.C. Davis**. (2023). *Epilogue: The Computer Engineer as Tool-User*. In Y.H. Lu & G.K. Thiruvathukal, *Intermediate C Programming* (2nd edition, pp. 439–442). CRC Press.

PATENTS

- [Pa-1] W. Jiang, C. Cheung, H. Kim, M. Kim, and **J.C. Davis**. *A method for identifying naming mismatches in neural networks based on their architectural properties*. Purdue University, U.S. provisional patent application 63/813,549. Filed May 28, 2025.
- [Pa-2] **J.C. Davis** and W. Davis. *Determining a validity of an event emitter based on a rule*. IBM, US Patent 11,875,185 B2. Granted Jan. 16, 2024.
- [Pa-3] W. Davis and **J.C. Davis**. *Verification of the Integrity of Data Files Stored in Copy-on-Write (CoW) Based File System Snapshots*. IBM, U.S. patent 11,176,090 B2. Granted Nov. 16, 2021.
- [Pa-4] **J.C. Davis** and W. Davis. *Injection of Simulated Hardware Failure(s) in a File System for Establishing File System Tolerance-to-Storage-Failure(s)*. IBM, U.S. patent 11,023,341 B2. Granted Jun. 1, 2021.
- [Pa-5] **J.C. Davis**, L. Rupprecht, D. Bhagwat, C. Arnold, and W. Sawdon. *Performing Hierarchical Provenance Collection*. IBM, U.S. patent 10,891,174 B1. Granted Jan. 12, 2021.
- [Pa-6] **J.C. Davis**, W. Davis. *File Metadata Verification in a Distributed File System*. IBM, U.S. patent 10,678,755 B2. Granted Jun. 9, 2020.
- [Pa-7] W. Davis and **J.C. Davis**. *Testing of Lock Managers in Computing Environments*. IBM, U.S. patent 10,061,777 B1. Granted Aug. 28, 2018.
- [Pa-8] **J.C. Davis**, W. Davis, and F. Knop. *Detection of File Corruption in a Distributed File System*. IBM, U.S. patent 10,025,788. Granted Jul. 17, 2018.

RESEARCH TRANSLATION AND IMPACT

SOFTWARE DEFECTS AND VULNERABILITIES

- [I-1] **1,098 confirmed software defects and vulnerabilities discovered.** Findings span major software infrastructure and open-source ecosystems, including Python, Node.js, Django, FreeRTOS, Zephyr, Contiki-NG, RIOT, NuttX, Pacman, npm packages, STM32 drivers, and hundreds of open-source embedded projects. Affected projects and services include ones maintained by Microsoft, Amazon, GitHub, and the Apache Foundation.
- [I-2] **24 assigned CVEs.** CVE-2018-7158, CVE-2018-1060, CVE-2018-1061, CVE-2018-7536, CVE-2023-34100, CVE-2023-34101, CVE-2023-37459, CVE-2023-37281, CVE-2023-35847, CVE-2023-35846, CVE-2023-35849, CVE-2023-35848, CVE-2024-41126, CVE-2024-41125, CVE-2024-38373, CVE-2025-5688, CVE-2025-9558, CVE-2025-9557, CVE-2025-1675, CVE-2025-1674, CVE-2025-1673, CVE-2026-5071, CVE-2026-25139, and CVE-2026-5072.

INDUSTRY AND OPEN-SOURCE ADOPTION

- [I-3] **Socket.** A research system is deployed in production at Socket, Inc., where it has identified more than 600 malicious packages across major software package registries.
- [I-4] **Google OSS-Fuzz-Gen.** Findings from research on LLM-assisted fuzzing were incorporated into Google's OSS-Fuzz-Gen to reduce false positives.
- [I-5] **AWS CBMC.** AWS changed defaults in its CBMC bounded-model-checking tool to expose latent vulnerabilities.
- [I-6] **Ruby.** The Ruby programming language adopted our ReDoS-mitigation technique.
- [I-7] **Node.js.** Research findings led to a security-guidance article on the Node.js website, security notes in documentation for several critical APIs, and a patch to the `fs.readFile` API.
- [I-8] **IBM.** GraphQL query cost analysis is incorporated into IBM API Connect's DataPower API Gateway.
- [I-9] **AJV.** Contributed a security defense to the widely used AJV JSON-schema validation library, which is used downstream by organizations including Microsoft and Mozilla.

RESEARCH SOFTWARE IN USE

- [I-10] **DocAble** (2026–present). Automated accessibility-remediation platform aligned with U.S. Department of Justice accessibility requirements. The system comprises more than 2 million lines of code and is used by Purdue Engineering to meet its ADA obligations. See <https://scholaccess.com>.
- [I-11] **safe-regex** (2018–present). Cyberinfrastructure for the JavaScript ecosystem that detects potentially unsafe regular expressions and helps identify regular-expression denial-of-service vulnerabilities. Directly used by 884 other npm packages and downloaded approximately 50 million times per month. See <https://www.npmjs.com/package/safe-regex>.

PUBLIC DISSEMINATION

- [I-12] **MITRE.** Research papers are cited by MITRE as further reading for CWE-1333, “Inefficient Regular Expression Complexity” (ReDoS).

- [I-13] ***The Economist***. A WWW’26 paper was featured in a research digest by *The Economist*.
- [I-14] ***IEEE Software: Practitioner’s Digest***. Two research works were selected for inclusion in the *IEEE Software* “Practitioner’s Digest” column.
- [I-15] **Practitioner-facing research writing**. Research summaries written for practitioners have received more than 80,000 views.

REFEREED MAGAZINE ARTICLES

- [M-1] T.R. Schorlemmer, E. Burmane, K. Kalu, S. Torres-Arias, and **J.C. Davis**. *Establishing Provenance Before Coding: Traditional and Next-Gen Software Signing*. IEEE Security & Privacy Magazine, special issue “Secure Software Before Codeing”, 2025 (IEEE S&P Magazine’25). 8 pages.
- [M-2] X. Hu, Z. Jiao, A. Kocher, Z. Wu, J. Liu, **J.C. Davis**, G.K. Thiruvathukal, and Y.H. Lu. *Evolution of Winning Solutions in the 2021 Low-Power Computer Vision Challenge*. IEEE Computer, 2023 (Computer’23). 6 pages.
- [M-3] A. Goel, C. Tung, N. Eliopoulos, A. Wang, **J.C. Davis**, G.K. Thiruvathukal, and Y.H. Lu. *Tree-based Unidirectional Neural Networks for Low-Power Computer Vision*. IEEE Design & Test, 2022 (IEEE D&T’22). 6 pages.

SELECTED OTHER REFEREED WORKS: SHORT PAPERS, PRELIMINARY RESULTS, ETC.

- [W-1] T. Singla and **J.C. Davis**. *Software Supply Chains are Dead: Use-Case-Oriented Regeneration*. Proceedings of the 20th International Symposium on Empirical Software Engineering and Measurement – Emerging Results, Vision, and Reflection Track) (ESEM-ERVR’26). 12 pages.
- [W-2] P. Amusuo, D. Liu, R.A.C. Méndez, J. Metzman, O. Chang, and **J.C. Davis**. *Lessons from Mitigating False Positives in Google’s OSS-Fuzz-Gen*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE) (FSE-Industry’26). 39% acceptance rate (66/171). 5 pages.
- [W-3] E. Pautsch*, T. Singla*, P. Kumar, W. Jiang, H. Peng, B. Hassanshahi, K. Läufer, G.K. Thiruvathukal, and **J.C. Davis**. *AgentHub: A Registry for Discoverable, Verifiable, and Reproducible AI Agents*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). Recommended for journal extension; 30/79 papers. (ICSE-JAWs’26). 9 pages.
- [W-4] H. Peng, P.V. Patil, A.Z. Qiu, G.K. Thiruvathukal, and **J.C. Davis**. *Beyond Local Code Optimization: Multi-Agent Reasoning for Software System Optimization*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering) (ICSE-JAWs’26). 5 pages.
- [W-5] H. Peng, A. Zhong, R.A.C. Méndez, K.G. Kalu, and **J.C. Davis**. *How Do Agents Perform Code Optimization? An Empirical Study*. Proceedings of the 23rd International Mining Software Repositories Conference — Mining Challenge track (MSR-Mining’26). 53% acceptance rate (62/116). 5 pages.
- [W-6] P.C. Amusuo, P.V. Patil, O. Cochell, T. Le Lievre, and **J.C. Davis**. *A Unit Proofing Framework for Code-level Verification: A Research Agenda*. Proceedings of the ACM/IEEE 47th International Conference on Software Engineering — New Ideas and Emerging Results track (ICSE-NIER’25). 26% acceptance rate (25/97). 5 pages.

- [W-7] P.V. Patil, W. Jiang, H. Peng, D. Lugo, K.G. Kalu, J. LeBlanc, L. Smith, H. Heo, N. Aou, and **J.C. Davis**. *Recommending Pre-Trained Models for IoT Devices*. Proceedings of the 7th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT'25). 5 pages.
- [W-8] S. Joshi, P. Mukherjee, K.A. Davis, and **J.C. Davis**. *Introducing Systems Thinking as a Framework for Teaching and Assessing Threat Modeling Competency*. Proceedings of the 2024 Annual Conference and Exposition of the American Society for Engineering Education (ASEE'24). ~50% acceptance rate (12 papers presented). 31 pages. *Best Paper Award, Software Engineering Division (given to 1 out of 12 papers)*.
- [W-9] **J.C. Davis**, P. Jajal, W. Jiang, T.R. Schorlemmer, N. Synovic, and G.K. Thiruvathukal. *Reusing Deep Learning Models: Challenges and Directions in Software Engineering*. Proceedings of the IEEE John Vincent Atanasoff Symposium on Modern Computing (JVA'23). 12 pages.
- [W-10] W. Jiang*, N. Synovic*, P. Jajal, T.R. Schorlemmer, A. Tewari, B. Pareek, G.K. Thiruvathukal, and **J.C. Davis**. *PTMTorrent: A Dataset for Mining Open-source Pre-trained Model Packages*. Proceedings of the 20th Annual Conference on Mining Software Repositories — Data and Tool Showcase Track (MSR-Data'23). 54% acceptance rate (23/42). 5 pages.
- [W-11] W. Jiang, N. Synovic, R. Sethi, A. Indarapu, M. Hyatt, T.R. Schorlemmer, G.K. Thiruvathukal, and **J.C. Davis**. *An Empirical Study of Artifacts and Security Practices in the Pre-trained Model Supply Chain*. Proceedings of the 1st ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED'22). 57% acceptance rate (12/21). 10 pages.
- [W-12] D. Montes, P. Peerapatanapokin, J. Schultz, C. Guo, W. Jiang, and **J.C. Davis**. *Discrepancies among Pre-trained Deep Neural Networks: A New Threat to Model Zoo Reliability*. Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering — Ideas, Visions, and Reflections track (ESEC/FSE-IVR'22). 25% acceptance rate (7/28). 5 pages.
- [W-13] C. Okafor*, T.R. Schorlemmer*, S. Torres-Arias, and **J.C. Davis**. *SoK: Analysis of Software Supply Chain Security by Establishing Secure Design Properties*. Proceedings of the 1st ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED'22). 57% acceptance rate (12/21). 10 pages.
- [W-14] N. Gopalakrishna, D. Anandayuvraj, A. Detti, F. Bland, S. Rahaman, and **J.C. Davis**. *“If security is required”: Engineering and Security Practices for Machine Learning-based IoT Devices*. Proceedings of the 4th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT'22). 8 pages.
- [W-15] **J.C. Davis**. *On the Impact and Defeat of Regex DoS*. ACM Student Research Competition, 2019–2020 Grand Finals. *Second place, graduate student division*.

SELECTED TECHNICAL REPORTS

- [R-1] **J.C. Davis**, K. Kalu, H. Peng, and P.V. Patil. *Model-Based Agentic Software Engineering*. <https://arxiv.org/pdf/2608.25174>. 2026.
- [R-2] **J.C. Davis**, P. Amusuo, T. Singla, B. Çakar, and K.A. Davis. *Cheap Code, Costly Judgment: A Case Study on Governable Agentic Software Engineering*. <https://arxiv.org/pdf/2607.01087>. 2026.

- [R-3] D. Lugo and J.C. Davis. *Is US Defense Acquisition Ready to Acquire AI-Enabled Capabilities? Assessing the DoD Software Acquisition Pathway Through a Scenario-Based Policy Analysis*. <https://arxiv.org/pdf/2606.07393>. 2026.
- [R-4] T. Sivertsen, N. Singh, and J.C. Davis. *SAGE: Structured Agentic Graph Editing for Software Diagrams*. <https://arxiv.org/pdf/2607.01102>. 2026.
- [R-5] S. Muralee, S. Cherupattamoolayil, J.C. Davis, A. Bianchi, and A. Machiry. *Reactive Bottom-Up Testing*. <https://arxiv.org/pdf/2509.03711>. 2025.
- [R-6] J.C. Davis, S. Chen, H. Peng, P. Amusuo, and K. Kalu. *A Guide to Stakeholder Analysis for Cybersecurity Researchers*. <https://arxiv.org/pdf/2508.14796>. 2025.
- [R-7] V. Banna, A. Chinnakotla, Z. Yan, A. Vegesana, N. Vivek, K. Krishnappa, W. Jiang, Y.H. Lu, G.K. Thiruvathukal, and J.C. Davis. *An Experience Report on Machine Learning Reproducibility: Guidance for Practitioners and TensorFlow Model Garden Contributors*. <https://arxiv.org/abs/2107.00821>. 2021.

COURSES DESIGNED* OR RE-DESIGNED†

†ECE 461 – Software Engineering <i>Purdue University</i>	Launched Fall 2021
*ECE 595 – Advanced Software Engineering <i>Purdue University</i>	Launched Spring 2021
†ECE 30862 – Software Engineering Tools <i>Purdue University</i>	Revamped Fall 2021

SELECTED COURSES TAUGHT

ECE 30861/461 – Software Engineering <i>Purdue University</i>	F21, S23, F23, F24, F25, F26
ECE 50874/59595 – Advanced Software Engineering <i>Purdue University</i>	S21, S22, S24, S26
ECE 368 – Data Structures <i>Purdue University</i>	Fall 2020
Vertically Integr. Project: Software Engineering w/Pre-Trained Models <i>Purdue University</i>	S24, F24, S25, F25, S26
Vertically Integr. Project: Open-Source TensorFlow Software <i>Purdue University</i>	F20, S21, F21, S22, F22, S23, F23

GRADUATE STUDENT ADVISING

DOCTORAL STUDENTS, AS COMMITTEE CHAIR

Dr. Wenxin Jiang	PhD	Graduated 2025
Dr. Paschal Amusuo	PhD	Graduated 2026
Dharun Anandayuvraj	PhD candidate	Fall 2021–present, expected Fall 2026

Purvish Jajal (with Y.H. Lu)	PhD candidate	Fall 2022–present, expected Fall 2027
Nicholas J. Eliopoulos (with Y.H. Lu)	PhD candidate	Fall 2024–present, expected Spring 2027
Kelechi Gabriel Kalu	PhD candidate	Spring 2023–present, expected Spring 2027
Daniel Lugo, US Space Force	PhD candidate	Fall 2024–present, expected Summer 2027
Andrew Rozema	PhD student	Fall 2024–present, expected Spring 2028
Berk Çakar	PhD student	Fall 2024–present, expected Spring 2029
Huiyun Peng	PhD student	Fall 2024–present, expected Spring 2029
Ricardo Andrés Calvo Méndez	PhD student	Fall 2025–present, expected Spring 2030

MSC-THESIS STUDENTS, AS COMMITTEE CHAIR

Tanmay Singla	MSc	<i>Graduated 2026</i>
Sofia Okorafor, US Navy	MSc	<i>Graduated 2026</i>
Parth V. Patil	MSc	<i>Graduated 2025</i>
Jason Jones	MSc	<i>Graduated 2024</i>
Taylor Schorlemmer, US Army	MSc	<i>Graduated 2024</i>
William Maxam, US Coast Guard	MSc	<i>Graduated 2023</i>
Geoffrey Cramer	MSc	<i>Graduated 2023</i>

Sammie Oviri Jim Okodaso	MSc	<i>Fall 2025–present, expected Spring 2027</i>
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DOCTORAL STUDENTS, AS COMMITTEE MEMBER

Dr. Qiang Xu	PhD	<i>Graduated 2024</i>
Dr. Abhinav Goel	PhD	<i>Graduated 2022</i>

Andreas Kellas (Columbia University)	PhD candidate	Spring 2025–present
Benjamin Shiue-Hal Chou	PhD candidate	Fall 2023–present
Chinenye Okafor	PhD candidate	Fall 2022–present

Sai Ritvik Tanksalkar	PhD student	Fall 2022–present
Akul Pillai	PhD student	Fall 2022–present
Mingjie Shen	PhD student	Fall 2021–present

MSC-THESIS STUDENTS, AS COMMITTEE MEMBER

Jayashree Srinivasan	MSc	<i>Graduated 2023</i>
Xiao Hu	MSc	<i>Graduated 2022</i>

STUDENT AWARDS AND FELLOWSHIPS

These are students I mentored.

NATIONAL AND INTERNATIONAL AWARDS

Huiyun Peng, “Noteworthy Reviewer” for USENIX Security 2026—Artifact PC	2026
Paschal C. Amusuo & Dharun Anandayuvraj, Qualcomm Innovation Fellows	2025
Evan Williams, CSGrad4US Fellow	2025
Arjun Gupte, Astronaut Scholar	2025
Dharun Anandayuvraj, Chateaubriand Fellow	2024
Matthew Hyatt, DOD NDSEG Fellow; NSF Graduate Research Fellow (GRFP)	2023
Vishnu Banna, Honorable Mention in CRA Undergraduate Research Competition	2021
Vishnu Banna, Google Open Source Peer Bonus	2021
Anirudh Vegesana, Google Open Source Peer Bonus	2021
Akhil Chinnakotla, Google Open Source Peer Bonus	2021

INSTITUTIONAL AWARDS

VIP Team: Presentation with Distinction Award, 2026 Spring Undergraduate Research Conference	2026
Arjun Gupte, VIP Outstanding Leadership Award	2026
Paschal C. Amusuo, CERIAS Diamond Award	2026
Paschal C. Amusuo, ECE Magoon Graduate Student Research Excellence Award	2026
Arjun Gupte, ECE Undergraduate Research Award	2026
Elijah Jorgensen, Rolls Royce Fellow	2026
Wenxin Jiang, ECE Magoon Graduate Student Research Excellence Award	2025
Jason Jones, ECE Magoon Graduate Student Teaching Excellence Award	2024
Paschal C. Amusuo, ECE Magoon Graduate Student Teaching Excellence Award	2022
Kyle Robinson, ECE Undergraduate Research Award	2024
Efe Barlas, ECE Undergraduate Research Award	2022
Vishnu Banna, ECE Undergraduate Research Award	2021
Joseph Bushagour, ECE Undergraduate Excellence Award for Teaching	2022
Joseph Thomas, ECE Magoon Graduate Student Teaching Excellence Award	2021
Venkat Manam, ECE Magoon Graduate Student Teaching Excellence Award	2021

SELECTED INVITED TALKS

ON MODEL-BASED AGENTIC SOFTWARE ENGINEERING (MAGE)

U.S. Space Force. “Brain Day”, unit Systems Delta (SYD) 85	2026
IBM T.J. Watson Research Center. Yorktown Heights, NY	2026
Argonne National Laboratories. Lemont, IL	2026
Purdue University, College of Engineering. ThinkAI Summer Workshop Series	2026
Purdue University, ECE. Workshop on “ <i>Job Interviews in the AI-Era</i> ”	2026

ON SOFTWARE ENGINEERING FOR MACHINE LEARNING — REUSING PRE-TRAINED MODELS

Oregon State University. Corvallis, OR	2025
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McGill University. Montreal, Canada	2025
Concordia University. Montreal, Canada	2025
The University of Arizona. Tucson, AZ	2024
Michigan Technological University. Houghton, MI	2024
Carnegie Mellon University. Pittsburgh, PA	2023
The University of Notre Dame. South Bend, IN	2023
Argonne National Laboratories. Lemont, IL	2023
ONNX Community Meetup, NVIDIA Headquarters. Santa Clara, CA	2023
Loyola University Chicago. Chicago, IL	2022

ON SOFTWARE ENGINEERING FOR EMBEDDED SYSTEMS

Securing Software Implementations through System Fuzz Testing and Modular Formal Methods <i>Rolls Royce Annual Conference, Cyber Technology Research Network. Indianapolis, IN</i>	2025
Quick and Dirty or Slow and Steady? Two Techniques for Validating Embedded Software <i>Rolls Royce Headquarters. Derby, UK</i>	2024

SELECTED ACADEMIC COMMUNITY SERVICE

EDITORIAL SERVICE

Co-Editor, “Practicing AI,” <i>IEEE Computer</i>	2026–present
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REVIEWER: MAJOR CONFERENCE TECHNICAL TRACKS⁵

Vice Chair, USENIX Security	2027
PC Member, ICSE	2025, 2026, 2027
PC Member, AAAI	2026, 2027
PC Member, USENIX Security	2025, 2026
PC Member, ESEC/FSE	2023, 2025
PC Member, ASE	2021, 2024
PC Member, ISSTA	2024, 2025

REVIEWER: JOURNALS

Reviewer, IEEE Transactions on Software Engineering (TSE)
Reviewer, ACM Transactions on Software Engineering and Methodology (TOSEM)
Reviewer, Springer Empirical Software Engineering (EMSE)
Reviewer, Journal of Systems and Software (JSS)
Reviewer, Journal of Online Trust & Safety (JOTS)

WORKSHOP AND EVENT ORGANIZATION

Organizer, Workshop on Using Generative AI as an Engineering Educator, ASEE 2026	2026
Co-Chair, ICSE Student Mentoring Workshop (ICSE-SMeW)	2024–2025

⁵In my research areas, service on the technical program committees (“PC member”) is a respected form of peer review service. This involves the review of 5–15 full papers per conference.

Panelist and Mentor, ICSE Student Mentoring Workshop (ICSE-SMeW)	2024
Mentor, ICSE Student Mentoring Workshop (ICSE-SMeW)	2023

NATIONAL SERVICE

US National Science Foundation, Panelist	2026
US National Science Foundation, Panelist	2025
US National Science Foundation, Panelist	2023

SELECTED UNIVERSITY SERVICE

ECE Representative on the COE AI in Education Committee	2026
ECE Representative on the COE Graduate Education Award Selection Committee	2025
Lead for BSc.-Computer Engineering, Purdue ECE ABET Committee 5-year self-evaluation	2024–2025
Member, Committee to Create MSc-Software Engineering	2024
Member of search committees	2022, 2023
Host, Purdue Engineering Distinguished Lecture Series (PEDLS) — Dr. Nancy Leveson (MIT)	2022

PROFESSIONAL MEMBERSHIPS

Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, Association for Computing Machinery (ACM)
Member, American Society for Engineering Education (ASEE)

SELECTED AWARDS AND RECOGNITION

FOR RESEARCH

NSF CAREER Award	2026
Paul C. Krause Faculty Research Award (<i>ECE@Purdue Junior Faculty Research</i>)	2025
Best Artifact Award, CCS'25 (PickleBall)	2025
Best Paper Award, Software Engineering Division, ASEE 2024 (<i>Educational research</i>)	2024
ACM Distinguished Paper Award, ESEC/FSE 2020	2020
Second place, Grand Finals of the ACM Graduate Student Research Competition	2020
First place, Graduate Student Research Competition, ESEC/FSE 2019	2019
ACM Distinguished Paper Award, ASE 2019	2019
Microsoft Security Researcher Acknowledgments (<i>for Regex DoS discoveries</i>)	2018
IBM Significant Contributor Award (<i>for Node.js</i>)	2018
ACM Distinguished Paper Award, ESEC/FSE 2018	2018
Pratt and Davenport Fellowships, Virginia Tech College of Engineering	2017–2020

FOR TEACHING

Purdue University Exceptional Early Career Teaching Award	2026
Purdue College of Engineering Faculty Excellence Award in Exceptional Early Career Teaching	2026
2022 Ruth and Joel Spira Outstanding Teacher Award	2022
“Outstanding Engineering Teachers” (COE) (<i>course evaluation scores</i>)	F'21, S'22, F'22, S'24

FOR SERVICE

ICSE 2026 Distinguished Reviewer Award	2026
FSE 2025 Distinguished Reviewer Award	2025
USENIX 2025 Distinguished Reviewer Award	2025
ASE 2021 Distinguished PC Member Award	2021
Outstanding Graduate Student Service Award, CS@VT	2020

FOR MENTORING

VIP Outstanding Team Mentor Award, Pre-Trained Models Team	2026
Outstanding Faculty Mentor — Purdue's Elmore School of Electrical & Computer Engineering	2024
VIP Outstanding Team Mentor Award, Purdue TensorFlow Team	2021

OTHER HONORS

Purdue ThinkAI Competition Winner (<i>COE-wide competition for creative use of GenAI</i>)	2026
Frederica Clarkson Award (<i>university-wide award for top graduating senior</i>)	2012